

TRIAI x Google DeepMind's AI Research foundations course

Challenge 3: Impact Statement Card

Project title: Early Detection of Toxic Language through Ubuntu-Aligned Latent Probing of LLMs

Participants of the project

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Large language models (LLMs) such as Gemma are increasingly embedded in everyday applications, from conversational agents to decision-support tools. Despite their transformative potential, these models remain vulnerable to producing racist, sexist, or otherwise toxic language. As Gehman et al. (2020) note, pre-trained large language models can easily degenerate into harmful outputs, creating risks of social harm, misinformation, and violations of ethical guidelines. Our project, *Early Detection of Toxic Language through Ubuntu-Aligned Latent Probing of LLMs*, seeks to address these challenges by embedding African ethical perspectives into toxicity detection methods.

Our focus is the use of classifier probes aligned with Ubuntu's value of relationality to detect toxicity at the hidden-state level of LLMs before harmful outputs are generated. This approach enhances safety and transparency, moving beyond opaque black-box filters that dominate current moderation systems. By incorporating African perspectives and synthetic Ubuntu-aligned datasets, our project ensures that toxicity detection reflects values of community, reciprocity, and interdependence rather than privileging Western individualistic norms. Over time, this framework can empower marginalized communities by embedding their ethical traditions into global AI systems, democratizing AI safety research and ensuring diverse voices shape responsible AI.

The societal impact of this project extends beyond technical safety. Grounding toxicity detection in Ubuntu philosophy emphasizes relational personhood, mutual responsibility, and the interdependence of individuals and communities. In this regard, hate speech is understood not merely as offensive language but as a rupture in social relationships. Thus, our project reframes toxicity detection as a safeguard for human dignity and social cohesion. This perspective aligns with values of fairness, autonomy, and collective well-being, ensuring that AI systems serve communities inclusively rather than reinforcing inequalities.

We acknowledge risks. Classifier probes may misclassify nuanced language, leading to false positives or negatives. Unequal adoption could persist if only well-resourced institutions integrate these methods, leaving vulnerable communities exposed. Bias may be reinforced if the dataset is not sufficiently representative. To mitigate these risks, we commit to iterative evaluation using accuracy, precision, recall, and F1-score.

Ultimately, our project responds to critical ethical questions: *whose preferences are LLMs aligned with, and whose perspectives shape toxicity detection?* By embedding Ubuntu's relational ethics, we ensure African voices are included in global AI safety discourse. This approach challenges Western dominance and demonstrates how African philosophy can guide responsible AI development. Looking forward, our Ubuntu-aligned probing framework offers a model for culturally grounded AI design that safeguards dignity, strengthens social cohesion, and ensures AI reflects the values of all humanity, not just a privileged subset

References

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